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Smart Parking detection with proximity recommendations

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1. Introduction

Finding a parking space quickly has become a major challenge in crowded areas. Drivers often spend a long time searching for available parking spots, wastes fuel and time, and negatively affects the overall driving experience. As the number of vehicles continues to increase, traditional parking management methods are becoming less efficient and less capable of handling real world parking demands.

1.1 Overview

Our project focuses on developing a smart parking system that utilizes computer vision and path-finding algorithms to improve parking efficiency in crowded environments. Therefore, the system is designed to automatically detect available parking spaces from visual data and guide drivers to the nearest empty spot. We aim to reduce the time spent searching for parking, decrease traffic congestion, and improve the overall driving experience.

1.2 Problem statement

Some people face difficulty finding a parking space in the shortest possible time, especially in crowded places such as malls and hospitals. Therefore, there is a need for an automated system capable of detecting the status of the nearest available parking space and enhancing the user's experience.

1.3 Goals

- 1- Detect and classify parking spaces as occupied or available.
- 2- Identify the nearest available parking space based on the user's location.
- 3- Guide drivers using the A* algorithm to find the optimal path.
- 4- Improve efficiency and reduce the time spent searching for parking spaces in crowded areas.

1.4 Gaps

Some systems rely on manual monitoring or sensor-based solutions, which may be costly, limited in scalability, or less effective in environments especially in real time. Therefore, many current systems only focus on detecting parking spaces without providing intelligent navigation to the nearest empty spot. There is also a lack of integrated solutions that combine real-time visual detection with effective path-finding algorithms to find the best path to guide drivers within parking facilities.

1.5 Proposed Solution

The project's novelty lies in the development of an automated, computer vision-based system that not only detects parking spaces in real time using visual data through a Yolo based object detection model but also intelligently guides drivers to the nearest available spot using shortest-path algorithms. Unlike traditional parking management solutions, the proposed system employs efficient visual analysis combined with spatial mapping and logic to detect parking availability in congested areas. The novelty stems from the integration of detection and navigation, adapting computer vision methods in a practical and scalable manner that minimizes the search time for a parking spot, making it highly suitable for real-world settings such as universities, hospitals, and shopping malls.

2. Related Work

Recent advances in computer vision, deep learning, and intelligent transportation systems have significantly improved parking occupancy detection and smart parking management solutions. Previous studies have explored different approaches ranging from traditional machine learning techniques to modern deep learning and IoT-based systems.

Tomikj and Kulakov [1] proposed a vehicle detection method based on Histogram of Oriented Gradients (HOG) combined with a Linear Support Vector Machine (SVM). Their approach focused on extracting handcrafted visual features from vehicle images to support object classification. Although the method demonstrated reliable performance with relatively low computational complexity, traditional HOG-based approaches remain sensitive to environmental variations such as illumination changes and shadows.

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Rajesh et al. [2] introduced an intelligent parking guidance system integrating computer vision, IoT, and edge computing technologies. Their system utilized deep learning techniques for parking occupancy detection and provided real-time parking recommendations to drivers. While the proposed framework achieved strong performance, the integration of IoT devices and edge computing components increased infrastructure complexity and deployment costs.

Amato et al. [3] presented a decentralized parking occupancy detection system using convolutional neural networks (CNNs). Their research demonstrated that deep learning approaches can significantly improve parking occupancy classification accuracy under varying environmental conditions. However, CNN-based systems typically require larger datasets and higher computational resources for effective training and deployment.

Another study investigated automated parking space detection using CNN-based classification models and fixed parking slot coordinates [4]. The proposed approach achieved accurate parking occupancy classification using the PKLot dataset. Nevertheless, the system depended on predefined parking slot configurations, limiting flexibility in dynamic parking environments where parking layouts or camera perspectives may change.

Hsu et al. [5] proposed a robust parking space detection system based on inter-space correlation analysis using traditional machine learning and probabilistic models. Their approach improved parking detection performance under challenging visual conditions; however, the multi-stage processing pipeline increased system complexity and computational overhead compared to modern end-to-end object detection architectures.

Srisura and Thakiguchi [6] proposed a Beacon proximity-based parking recommendation service designed for indoor parking environments where GPS signals are unreliable. Their system utilized Bluetooth Low Energy (BLE) Beacon devices and NFC-based authentication to improve parking recommendation accuracy. Although the system demonstrated improved indoor localization performance, it required additional hardware infrastructure and specialized device deployment.

Other studies have explored generalized parking occupancy analysis using deep convolutional neural networks and computer vision techniques [7]. These approaches achieved high classification accuracy but often relied on computationally intensive architectures that may limit real-time deployment efficiency in large-scale parking systems.

Research has also investigated parking occupancy detection using deep learning and computer vision frameworks for automated parking management [8]. These systems demonstrated promising occupancy classification results; however, several approaches required extensive preprocessing stages and large training datasets to achieve stable performance.

In addition, Liu et al. [9] investigated parking guidance systems based on computer vision for intelligent transportation applications. Their study focused on improving parking management and vacancy monitoring through image-based parking analysis. Despite the effectiveness of the proposed framework, the study provided limited emphasis on modern real-time object detection architectures and optimized inference performance.

In this project, YOLOv8 is adopted as the primary object detection architecture due to its balance between detection accuracy, real-time inference speed, and computational efficiency. Unlike several previous studies that relied on handcrafted feature extraction, fixed parking slot configurations, or complex IoT infrastructures, the proposed system utilizes an end-to-end deep learning pipeline for efficient parking occupancy detection and intelligent parking analysis.

3. Dataset and Preprocessing

The data set utilized in this project is primarily derived from the PKLot parking dataset. To reduce the time and effort required for manual annotation, a pre-labeled version of the same dataset was obtained from Roboflow Universe. This version includes ready-made annotations for parking-related objects and vehicles using bounding boxes in YOLO format.

The preprocessing stage is responsible for preparing and organizing the parking images before they are passed to the YOLOv8 model for training and inference. Proper preprocessing is essential to ensure dataset consistency and support efficient model learning.

The preprocessing pipeline includes the following steps:

Dataset Organization:

The dataset is organized into training, validation, and testing subsets to support model training and evaluation.

Data Cleaning:

The dataset and labels were manually checked to ensure consistency and correct formatting before training.

Image Resizing:

All images are automatically resized to 640×640 pixels to ensure compatibility with the YOLOv8 architecture and maintain consistent input dimensions across the dataset.

The overall objective of preprocessing is to provide clean, standardized, and properly annotated image data that enables accurate and efficient parking occupancy detection using YOLOv8.

4. Proposed Methodology and Implementation

A. Overall System Workflow

The proposed system integrates computer vision techniques with intelligent path-finding algorithms to automate parking management and assist users in finding the nearest available parking space. The system workflow is divided into two main stages:

1. Parking Detection and Classification Stage:

The **YOLOv8 model** is used to analyze parking lot images, detect parking spaces, and classify each parking space as either occupied or vacant.

2. Nearest Parking Recommendation Stage:

The **A* algorithm** uses the detected parking space information to calculate the shortest route from the user's current location to the nearest available parking space.

B. System Implementation Phases

1. Parking Space Detection and Classification Using YOLOv8

In this stage, the system processes parking lot images from the dataset using the **YOLOv8 model** to detect and classify parking spaces. The process is performed **through the following steps**:

- **Parking Space Detection**

The system receives a parking lot image and identifies the locations of parking spaces within the scene using object detection techniques.

- **Parking Space Classification**

After detecting the parking spaces, the **YOLOv8 model** analyzes each parking space and classifies it into one of the following categories:

- **Occupied Space:** A parking space containing a car.
- **Available Space:** A parking space that is empty and available.

The output results include bounding boxes around the detected parking spaces along with classification labels indicating whether each parking space is occupied or vacant.

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- **Detection Results Processing**

After the detection and classification process is completed, the system processes the detected parking spaces and extracts the parking spaces classified as vacant. The locations of the available parking spaces are then prepared to be used in the nearest parking space recommendation stage.

2. Proximity Recommendation Using the A* Algorithm

After identifying the vacant parking spaces, the **A* algorithm** is used to determine the shortest route from the user's location to the nearest available parking space through the following steps:

- **Parking Layout Representation**

The parking environment is transformed into a 2D grid representation to simplify path planning. **Within this grid:**

- Navigable areas and vacant parking spaces are treated as accessible paths.
- Occupied parking spaces and obstacles are treated as blocked areas that cannot be crossed.

- **Starting Point Identification and Search**

The algorithm begins by defining the user's current location as the starting point. It then searches through the accessible paths to identify the nearest vacant parking space.

- **Path Calculation and Routing**

The algorithm calculates the shortest path between the user's location and the selected parking space. Finally, the computed route is visualized on the parking image to guide the user toward the recommended parking space.

C. System Integration and Data Flow

The system is designed to ensure smooth integration between the **YOLOv8 detection model** and the **A* pathfinding algorithm**. The parking space detection results are directly used to update the parking layout representation, allowing the **A* algorithm** to calculate the shortest route based on the detected parking availability. This integration enables the system to generate accurate and efficient parking recommendations using the analyzed parking images. accurate, real-time status data extracted from the analyzed images.

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4.1 System Architecture

Our system is based on intelligent detection of available parking spaces combined with recommending the nearest parking spot. The system is designed as a sequential series of stages to ensure smooth data flow and allow future development and scalability.

The system begins with the parking data set stage, which contains images or video clips of parking lots captured under different conditions such as variations in lighting and camera angles. After that, the data is loaded into the working environment for processing.

Next, data cleaning is performed. In this stage, corrupted, distorted, duplicated, or low-quality images are removed to prevent negatively affecting the model's performance and maintaining dataset consistency before the training process. After that, image preprocessing is applied. All images are resized to a unified fixed size to ensure consistency across the dataset. The dataset is then split into training, validation, and testing sets. During model training and inference, the YOLO model performs internal processing such as normalization automatically; therefore, no additional manual normalization is applied in the preprocessing stage.

Then, the parking space detection stage is performed. In this stage, parking spaces are identified within the image or video using a YOLO object detection model to detect parking spaces and their occupancy status.

After extracting each parking slot individually, the system proceeds to the occupancy classification stage. In this stage, each parking space is analyzed to determine whether it is available or occupied based on the visual information extracted from the image.

Next, the system moves to the decision-making and routing stage. First, only the available parking spaces are filtered from the detected slots. The user's location is mapped directly onto the parking lot image as a visual reference point. To determine the optimal path within the parking layout, the system applies the A* search algorithm, which is used to compute the shortest and most efficient route from the user's position to the nearest available parking space based on a graph representation of the parking structure. The closest vacant slot is then highlighted visually on the interface as the optimal choice.

After selecting the parking space, the system determines the route within the same visual environment, guiding the user through the parking layout using image-based navigation enhanced by the computed A* path rather than external maps or numerical calculations.

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Finally, the results are displayed through the user interface. The parking lot image is shown with occupied and available spaces clearly distinguished, while both the user's position and the nearest available parking space are visually highlighted to provide clear guidance.

Overall, architectural design allows future scalability, such as integrating live cameras, which makes the system suitable for real-world applications.

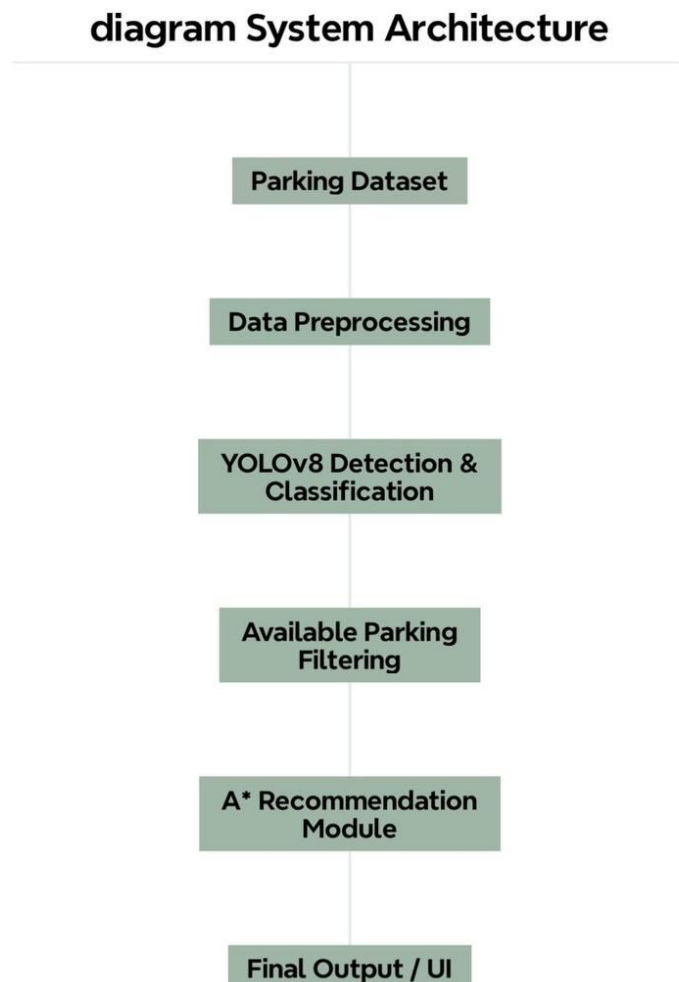


Figure 1: Proposed System Architecture

4.2 Parking Space Detection and Classification Using YOLO

First: Dataset Preparation

A parking dataset was downloaded, consisting of real-world images of different parking scenarios, **along with pre-labeled data (Labels)** that specify the state of each parking space within the image. The dataset includes **two main classes: space-empty and space-occupied**.

This data set serves as the foundation for training the model to learn patterns and accurately distinguish between different parking states.

Second: Model Training using YOLO

The YOLO model was used to train the system to detect and classify parking spaces. No separate preprocessing step was applied before training, as the model relies on its internal processing mechanisms during the training phase.

The training parameters were set as follows:

- **Epochs = 15:** The dataset was passed through the model 15 times, allowing it to gradually learn and improve its performance.
- **Batch Size = 8:** The model was trained on 8 images per iteration, achieving a balance between training speed and memory usage.
- **Image Size = 640:** The input images were set to 640×640 to meet the YOLO model requirements.

During the training process, the model gradually reduces the error value (Loss), which reflects the difference between predicted and actual results. **Through this process, the model learns how to:**

- Detect the location of parking spaces within the image (Object Detection).
- Classify each parking space as either empty or occupied (Classification).

Third: Model Evaluation

After completing the training process, the model was evaluated using the validation dataset. This evaluation aims to measure the model's ability to accurately detect and classify parking spaces.

Several performance metrics were used to assess the model's quality, including:

- **mAP50:** Measures the model's accuracy at an Intersection over Union (IoU) threshold of 0.5
- **mAP50–95:** Provides a more comprehensive evaluation across multiple IoU thresholds
- **Precision:** Measures the proportion of correct positive predictions out of all positive predictions
- **Recall:** Measures of the model's ability to detect all relevant instances

High values in these metrics indicate that the model is capable of effectively distinguishing between empty and occupied parking spaces.

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Fourth: Model Testing

The model was tested using new data (Test Data) that was not used during training in order to evaluate its ability to generalize to unseen data. The best trained model (**best.pt**) was loaded and used to perform predictions on the test images, with a **confidence threshold of 0.25** applied to ensure that only reliable predictions were considered.

The model analyzed each image, detected the locations of parking spaces, and classified each space as either Empty or Occupied. The prediction results were automatically saved in a designated folder, including images with bounding boxes and labels, for further analysis and presentation.

Fifth: Model Saving

The best trained model (**best.pt**) was saved to Google Drive to ensure it can be reused later without the need for retraining, whether for testing or future use.

Results and Output

The output results demonstrate the overall performance of the system across the training, evaluation, and testing stages. During training, the model showed a clear improvement over epochs, where the loss values gradually decreased, indicating that the model was learning effectively and improving its detection and classification capabilities.

In the evaluation stage, the model achieved very high-performance metrics, with **mAP50 reaching approximately 0.99** and **mAP50-95 around 0.95**. In addition, the **Precision reached approximately 0.997** and the **Recall reached approximately 0.997**, indicating that the model is highly accurate and capable of correctly detecting and classifying parking spaces.

In the testing stage, the model was applied to unseen images, where it **successfully detected parking spaces and classified them as either empty or occupied**. The output images clearly show bounding boxes around each parking space along with confidence scores, reflecting the reliability of the predictions.

Overall, the results confirm that the model performs effectively across different scenarios and demonstrates strong generalization ability.

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Screenshots of the outputs

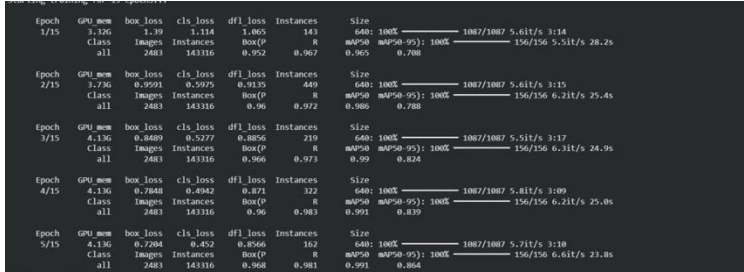


Figure 2: Training progress over epochs showing loss reduction.

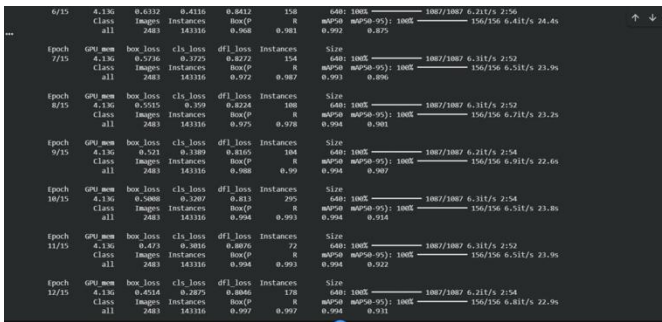


Figure3: Continued training results and performance improvement.

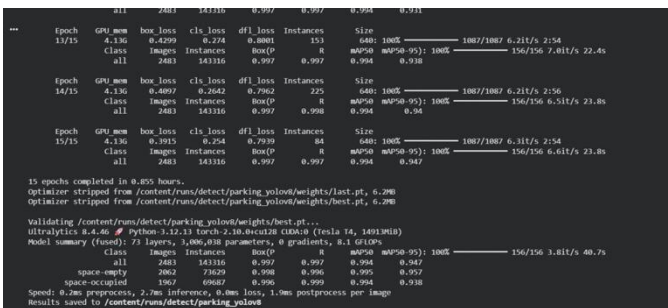


Figure 4: Final training stage and model performance metrics.

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```
# Run detection on test images
results = model.predict(
    source='/content/datasets/test/images',
    conf=0.25,
    save=True
)

Image 1187/1242 /content/datasets/test/images/2013-04-12 10 35 05 .jpg.rf.26c3d3db47ba6473d6b99673351a18.jpg: 640x640 40 space-occupied, 7.2ms
Image 1188/1242 /content/datasets/test/images/2013-04-12 10 45 06 .jpg.rf.c93694823b21f6423d3b86395346.jpg: 640x640 2 space-empty, 28 space-occupied, 8.5ms
Image 1189/1242 /content/datasets/test/images/2013-04-12 11 35 06 .jpg.rf.e25f3f5f713dbf6c5e81c3ab3a4232.jpg: 640x640 5 space-empty, 35 space-occupied, 7.2ms
Image 1190/1242 /content/datasets/test/images/2013-04-12 11 40 06 .jpg.rf.11a2ad712a92083978c996709673ff.jpg: 640x640 3 space-empty, 38 space-occupied, 8.3ms
Image 1191/1242 /content/datasets/test/images/2013-04-12 13 25 08 .jpg.rf.ab5d4c4423960f707f13d6b43419f.jpg: 640x640 11 space-empty, 29 space-occupied, 7.2ms
Image 1192/1242 /content/datasets/test/images/2013-04-12 14 50 09 .jpg.rf.1b506819866a6e57367a45aeb1edc.jpg: 640x640 40 space-occupied, 7.2ms
Image 1193/1242 /content/datasets/test/images/2013-04-12 16 30 11 .jpg.rf.15072406097f46e7a1ed41e27ab394a.jpg: 640x640 1 space-empty, 39 space-occupied, 7.1ms
Image 1194/1242 /content/datasets/test/images/2013-04-12 16 50 11 .jpg.rf.12ed5c9d6ac9eb0ab0a3db9513.jpg: 640x640 6 space-empty, 24 space-occupied, 7.2ms
Image 1195/1242 /content/datasets/test/images/2013-04-13 06 50 01 .jpg.rf.ab54ace0c977c2e7a5a8f92b809eeeb.jpg: 640x640 40 space-empty, 7.0ms
Image 1196/1242 /content/datasets/test/images/2013-04-13 07 50 02 .jpg.rf.f018b53b21a6fca8ff23d992d65b7.jpg: 640x640 40 space-empty, 8.1ms
Image 1197/1242 /content/datasets/test/images/2013-04-13 07 55 02 .jpg.rf.bebd22839af2fd47a7972b64867.jpg: 640x640 40 space-empty, 8.2ms
Image 1198/1242 /content/datasets/test/images/2013-04-13 08 15 02 .jpg.rf.7b91b56d8c51f7b6ac0113d48972e04.jpg: 640x640 40 space-empty, 9.0ms
Image 1199/1242 /content/datasets/test/images/2013-04-13 09 25 03 .jpg.rf.ab621839d3c22215b0d956a7231455.jpg: 640x640 38 space-empty, 2 space-occupied, 7.2ms
Image 1200/1242 /content/datasets/test/images/2013-04-13 09 30 04 .jpg.rf.ab743ed1ef43f3a07002797c4c1a.jpg: 640x640 38 space-empty, 2 space-occupied, 7.1ms
Image 1201/1242 /content/datasets/test/images/2013-04-13 12 45 07 .jpg.rf.c05caf21b0e5f7d6c3a3a05946983.jpg: 640x640 24 space-empty, 16 space-occupied, 7.2ms
Image 1202/1242 /content/datasets/test/images/2013-04-13 13 55 08 .jpg.rf.47089a2a8b706b52646de91ffc61f.jpg: 640x640 25 space-empty, 15 space-occupied, 8.1ms
Image 1203/1242 /content/datasets/test/images/2013-04-13 14 25 09 .jpg.rf.4f3f9d6ecf7db82a79733ba0b713.jpg: 640x640 25 space-empty, 15 space-occupied, 7.2ms
Image 1204/1242 /content/datasets/test/images/2013-04-13 15 45 10 .jpg.rf.e43f98a13382932a7a88b5d5d41cb72.jpg: 640x640 30 space-empty, 10 space-occupied, 7.2ms
Image 1205/1242 /content/datasets/test/images/2013-04-13 15 55 11 .jpg.rf.1a381c291b3d38c7839957715f6b045c.jpg: 640x640 31 space-empty, 9 space-occupied, 7.2ms
Image 1206/1242 /content/datasets/test/images/2013-04-13 16 00 11 .jpg.rf.ad3a1c24f3409a63c15c1a0b0a8d5c.jpg: 640x640 31 space-empty, 9 space-occupied, 7.2ms
Image 1207/1242 /content/datasets/test/images/2013-04-13 16 35 11 .jpg.rf.735627e2b0fe99ad08f8c6d857cc04.jpg: 640x640 30 space-empty, 10 space-occupied, 9.9ms
Image 1208/1242 /content/datasets/test/images/2013-04-13 17 25 12 .jpg.rf.9d1121c8e5d0e725f8d725045f02a.jpg: 640x640 33 space-empty, 7 space-occupied, 7.2ms
```

Figure 5: Model prediction results on test images.

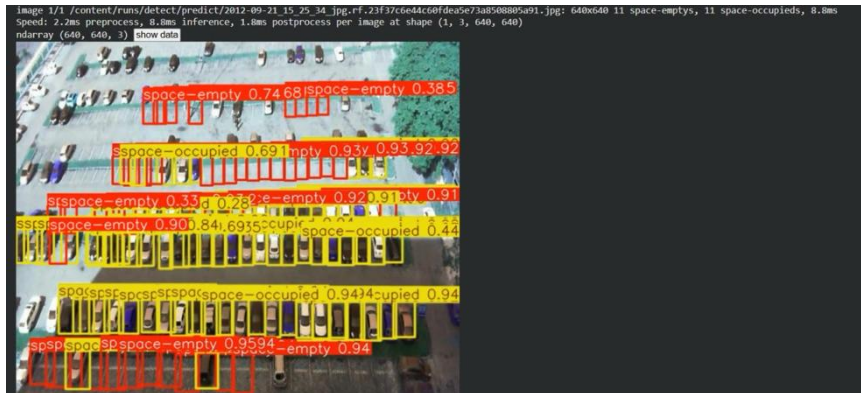


Figure 6: Detection results under different scenarios.



Figure 7: Detection results showing empty parking spaces.

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Figure 8: Detection of empty and occupied parking spaces.

```
# Evaluate the trained YOLOv8 model on the validation dataset
metrics = model.val()

# Print evaluation metrics to measure model performance
print("mAP50:", metrics.box.map50)
print("mAP50-95:", metrics.box.map)
print("Precision:", metrics.box.mp)
print("Recall:", metrics.box.mr)

Ultralytics 8.4.46 Python-3.12.12 torch-2.10.0+cu128 CUDA:0 (Tesla T4, 14913MiB)
Model summary (fused): 73 Layers, 3,806,038 parameters, 0 gradients, 8.1 GiB
val: Fast image access (ping: 0.010, 0 ms, read: 1700.51686.6 MB/s, size: 71.7 KB)
val: Scanning /content/datasets/valid/labels.cache... 2483 images, 59 backgrounds, 0 corrupt: 100% 2483/2483 650.9Mit/s 0.0s
Class Images Instances Box(P R mAP50 mAP50-95): 100% 156/156 3.9it/s 40.1s
space-empty 2483 143316 0.997 0.997 0.994 0.951
space-occupied 2062 73629 0.998 0.996 0.995 0.962
space-occupied 1967 69687 0.996 0.999 0.994 0.941
Speed: 1.2ms preprocess, 3.9ms inference, 0.0ms loss, 1.9ms postprocess per image
Results saved to /content/runs/detect/val
mAP50: 0.9943903997971385
mAP50-95: 0.9512612414038664
Precision: 0.997133264790302
Recall: 0.9971894912922228
```

Figure 9: Evaluation metrics

4.3 Nearest Parking Recommendation Algorithm

4.3.1 Overview

The system is based on intelligent detection of available parking spaces, combined with Recommending the nearest parking space. The system combines parking space detection, occupancy classification, and shortest-path computation to create an efficient and intelligent parking recommendation process. The A algorithm was chosen specifically because it is one of the most efficient and fast pathfinding algorithms for finding the shortest path. A uses a heuristic function to significantly reduce the number of paths explored, making it ideal for providing near-real-time navigation in parking environments.

The proposed system helps reduce time and effort and decreases traffic congestion, while also enhancing the drivers' experience by making the process of finding a parking spot faster and more efficient. It is very useful in real-world environments, especially in crowded places such as universities, hospitals, and malls.

4.3.2 System Workflow

The workflow below depicts the overall process of the proposed system, which includes image acquisition, parking space detection, pathfinding, and recommendation generation.

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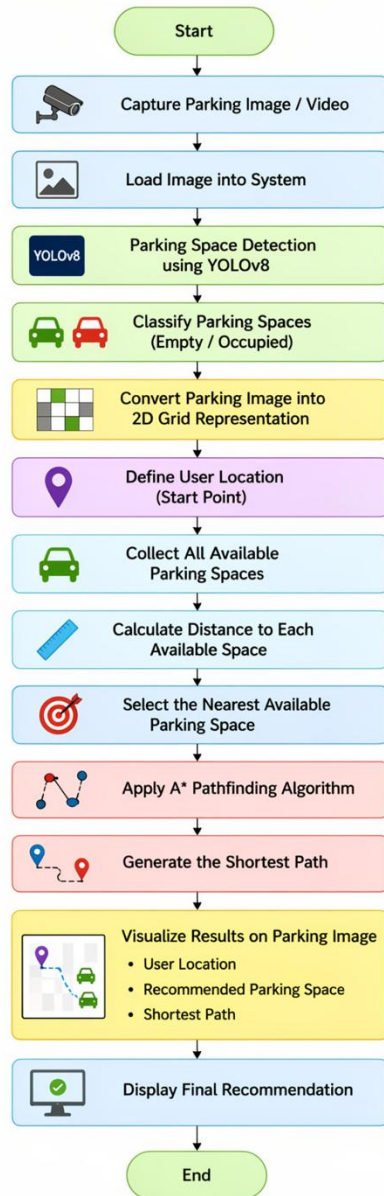


Figure 10: the complete workflow of the proposed system

4.3.3 Parking Recommendation Strategy and Pathfinding

The system first filters only the available parking spaces. It receives the user's location as input and calculates the distance between the user's position and each available parking space to select the nearest vacant spot based on the shortest distance. After selecting the parking space, the system determines the route to reach it using the A* search algorithm.

The parking environment is represented as a 2D grid where free cells represent navigable areas and blocked cells represent obstacles. The starting point is defined near the bottom of the image and converted into grid coordinates.

The A* algorithm is then applied to compute the shortest path between the user and the selected parking space by calculating the total path cost using the following equation:

$$f(n) = g(n) + h(n)$$

Where:

- $g(n)$ represents the actual path cost from the starting node.
- $h(n)$ represents the estimated remaining distance to the target node.
- $f(n)$ represents the total estimated cost.

Based on the computed paths, the system selects the parking space with the lowest path cost and recommends it to the driver. Finally, the parking lot image is shown with occupied and available spaces clearly distinguished, and the computed path is visualized on the image using plotted points to guide the driver along the estimated navigation route.

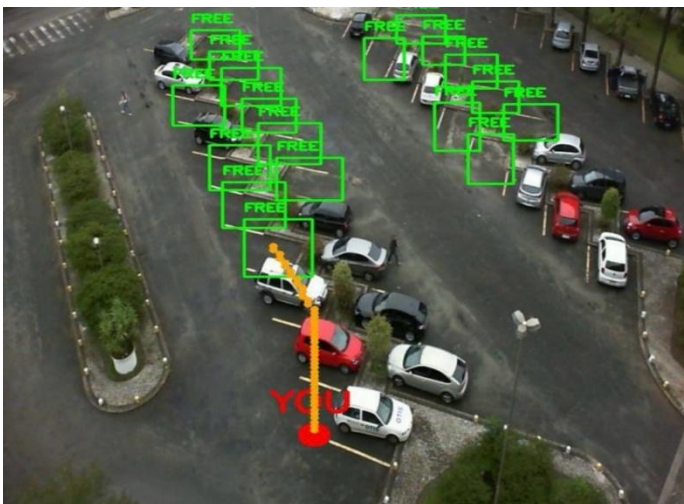


Figure 11: shortest path to the nearest available parking space

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4.4 Interactive User Interface

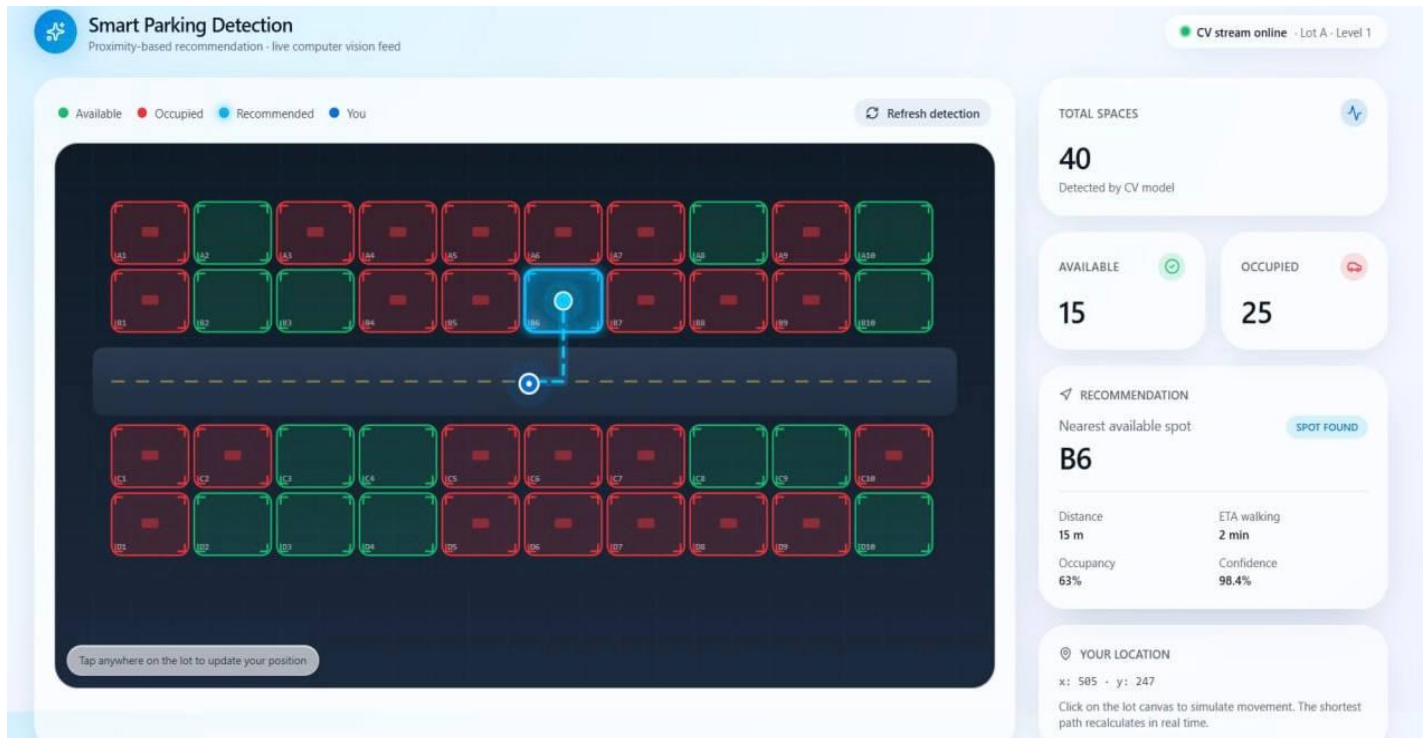


Figure 12: Interactive User Interface for Smart Parking Detection and Nearest Parking Recommendation

The developed user interface provides an interactive visualization of the proposed smart parking system. The interface displays parking spaces with different colors representing their status, where green indicates available spaces and red indicates occupied spaces. In addition, the user's current location is represented within the parking layout, while the recommended nearest parking space is highlighted separately. The system also visualizes the shortest calculated route between the user's location and the recommended parking space using the A* algorithm.

The interface supports dynamic interaction by allowing the user position to be changed within the parking layout. Whenever the user location changes or the detection process is refreshed, the system automatically recalculates the nearest available parking space and updates the displayed route accordingly. Furthermore, the interface includes statistical information such as the total number of parking spaces, available spaces, occupied spaces, and recommendation details to improve usability and system visualization

5. Experimental Results

Model Evaluation Metrics

Standard computer vision metrics such as Precision, Recall, F1-Score, and mean Average Precision (mAP) were used to thoroughly evaluate the performance of the proposed parking slot detection model. The experimental results show that the model achieves exceptional localization and classification accuracy while retaining robust generalization across both empty and occupied parking slot states.

Confusion Matrix Analysis

The absolute and normalized confusion matrices provide a granular view of the model's true versus predicted classifications:

- **Empty Parking Spaces (space-empty):**

The model correctly classified 73,219 samples, achieving a remarkable 99% true positive rate (Normalized CM = 0.99). Misclassifications were negligible; only 373 instances of actual empty spaces were falsely predicted as occupied, and a mere 37 instances were missed as background.

- **Occupied Parking Spaces (space-occupied):**

The model achieved a near-perfect classification rate of 1.00 in the normalized confusion matrix, successfully detecting 69,580 instances. Misclassifications were extremely rare, with only 107 instances falsely predicted as empty spaces.

- **Background Analysis:** The normalized matrix reveals that unmatched background regions were assigned to empty spaces and occupied spaces at rates of 54% and 46%, respectively. However, this background confusion has zero negative impact on the core task, as actual parking slots (both empty and occupied) demonstrate near-perfect sensitivity (99% and 100% respectively), ensuring high reliability during realtime deployment.

Performance Curves Analysis

The evaluation curves confirm the model's high operational stability across various confidence thresholds:

- **Precision-Recall (PR) Curve:** The model achieved an outstanding mean Average Precision of $mAP@0.5 = 0.994$ (99.4%) across all classes. Individually, the space-empty class scored a mAP of 0.995, and the space-occupied class scored 0.994. The PR curve remains close to the upper-right corner, signifying that the model maintains peak precision without sacrificing its recall rate.
- **F1-Confidence Curve:** The global F1-score reaches approximately 1.00 at a confidence threshold of 0.578. This represents the optimal operational point for deployment, perfectly balancing precision and recall.
- **Precision-Confidence Curve:** The precision remains exceptionally high (near 1.0) throughout almost the entire threshold spectrum, remaining consistently close to 1.00 across nearly all confidence thresholds.
- **Recall-Confidence Curve:** The model maintains a perfect recall rate (1.00) from a threshold of 0.000 up to approximately 0.80, proving that it successfully avoids missing parked vehicles or empty spots even at high confidence requirements.

Qualitative Evaluation (Visual Inspection)

A qualitative analysis of the model's performance on real-world surveillance imagery validates the quantitative metrics.

As shown in the detection batches, the model successfully overcomes environmental challenges such as tree foliage, casting shadows, and perspective distortions at the edges of the camera lens:

- Vacant spaces are accurately bounded in dark blue boxes (space-empty) with high confidence scores typically above 0.85.
- Occupied spaces are uniformly mapped using cyan boxes (space-occupied), tracking long rows of parked cars without getting confused by dense vehicle packing or partial occlusions.

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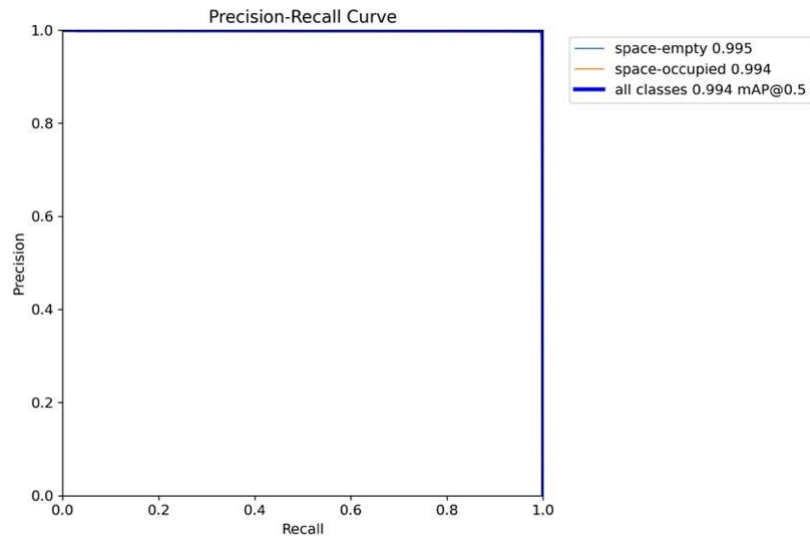


Figure 13: Precision-Recall performance of the proposed parking slot detection model.

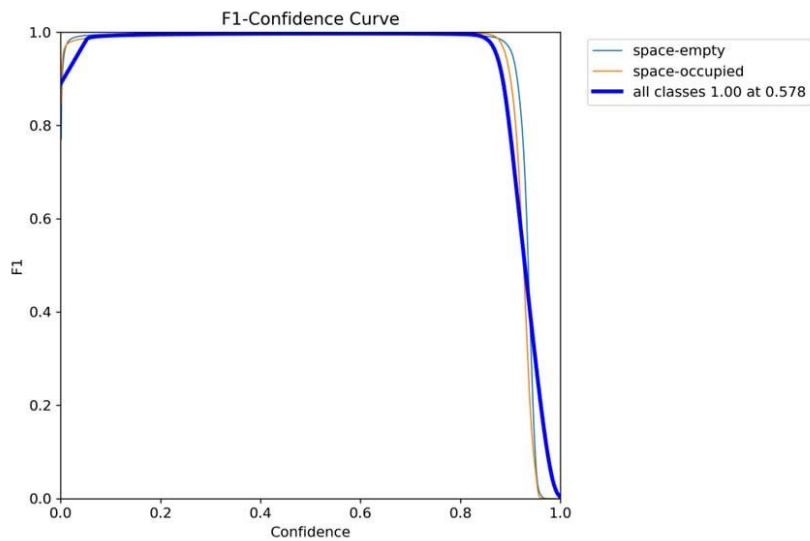


Figure 14: F1-score variation across different confidence thresholds.

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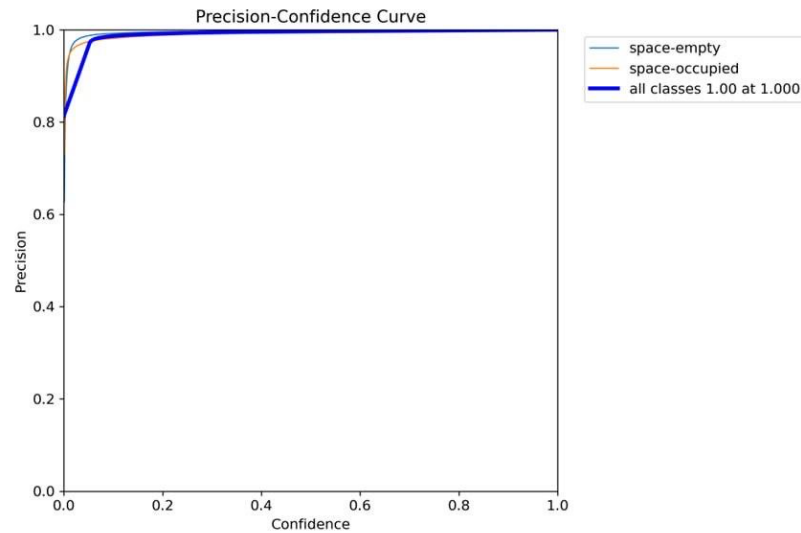


Figure 15: Precision performance under varying confidence thresholds.

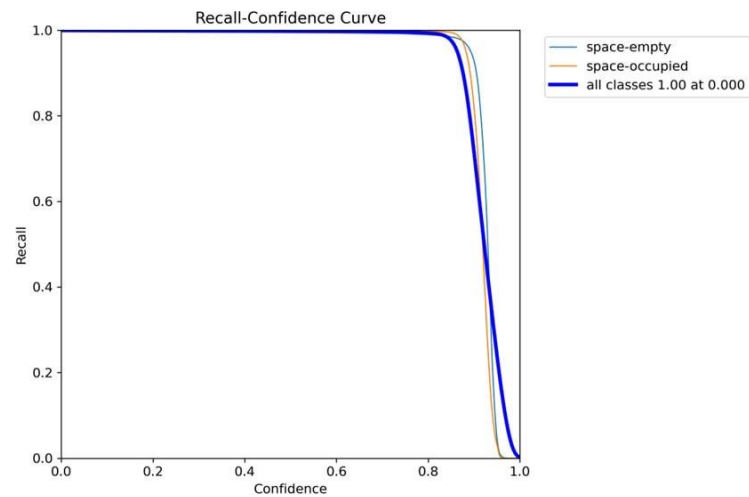


Figure 16: Recall performance across confidence threshold levels.

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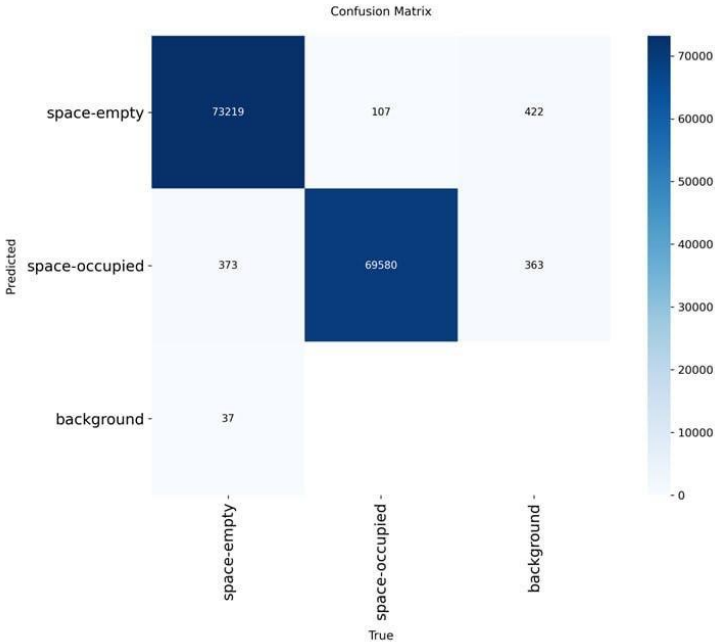


Figure 17: Absolute confusion matrix for parking slot classification results.

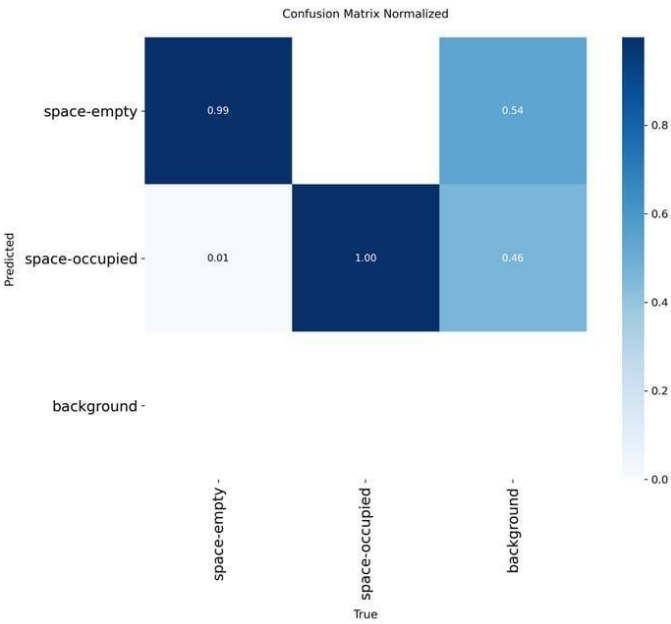


Figure 18: Normalized confusion matrix illustrating classification performance.

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Figure 19: Sample detection outputs under real-world parking conditions.

6. Discussion

The experimental results confirm the high effectiveness of integrating **YOLOv8**-based computer vision with the **A* path-finding algorithm** for smart parking management. The system's ability to achieve a high Precision and Recall (**approximately 0.997**) directly supports its reliability in reducing unnecessary vehicle movement and saving drivers' time.

A major strength identified is the seamless transition from occupancy detection to shortest-path recommendation using a 120 x 120 grid representation. This structured environment allowed the **A*** algorithm-selected for its efficient use of heuristics-to provide near real-time navigation.

Despite these strengths, performance can be influenced by environmental variables. Specifically, extreme lighting conditions or shadows may challenge the **YOLO** model's detection accuracy. Furthermore, if the 2D grid does not perfectly align with the physical environment, the **A* algorithm** might generate suboptimal paths.

Looking forward, the system's architecture is highly scalable, with future plans to integrate GPS and Google Maps for a seamless navigation experience from main roads to the parking spot. Expanding to live camera feeds and multi-story support will further adapt the system for large-scale facilities like universities and hospitals, aligning it with Smart City infrastructure goals.

7. Error Analysis

During the evaluation phase, the model demonstrated high detection accuracy; however, some errors occurred in complex scenarios such as low lighting and overlapping vehicles. Table 1 summarizes the observed errors and their possible causes.

Error Type	Description	Possible Cause	Impact
False Positive	Occupied space detected as empty	Shadows or reflections	Incorrect parking recommendation
False Negative	Empty space not detected	Low lighting or occlusion	Available parking may be missed
Misclassification	Wrong parking state classification	Similar visual patterns	Reduced detection accuracy
Bounding Box Inaccuracy	Incorrect object localization	Overlapping vehicles	Imprecise parking detection
Label Congestion & Visual Clutter	Overlapping labels and bounding boxes in dense areas.	Sharp camera angle and perspective projection in crowded sectors	Creates a crowded visual output but maintains accurate core classification.

Table 1: Detection error analysis.



Figure 20: Label congestion

This scene shows visual clutter caused by a sharp camera perspective. Despite the overlapping labels, the YOLOv8 model accurately classifies vacant slots (space-empty) and occupied zones (space-occupied) without losing detection accuracy.

8. Technical Challenges and Solutions

1. Total Failure of Traditional Detection Methods (HOG)

- **Challenge:** Initial attempts to implement the **HOG (Histogram of Oriented Gradients)** algorithm yielded no functional results. The algorithm failed to produce any detections or meaningful output when applied to the parking dataset, proving it completely incapable of handling the complexity of the task.
- **Solution:** Recognizing this total failure, the project immediately pivoted to a **Deep Learning** approach using **YOLOv8**. This shift was essential as it moved from a failing manual feature extraction method to a robust, automated neural network capable of processing the data successfully.

2. Hardware Constraints and Computational Demands

- **Challenge:** Training advanced computer vision models like YOLOv8 imposed a significant burden on the available hardware. The initial device lacked the necessary **GPU capabilities**, leading to excessive training times and extreme latency that hindered the development progress.
- **Solution:** To overcome this infrastructure bottleneck, the workload was migrated to a high-performance laptop equipped with an **NVIDIA GPU**. By leveraging **CUDA-accelerated** parallel processing, the training cycle was optimized.

3. Software Library Compatibility

- **Challenge:** Security updates in the software libraries introduced restrictive protocols that blocked the model from loading its pre-trained weights.
- **Solution:** This was resolved by implementing a **programmatic override**. A custom function wrapper was used to adjust the loading parameters, ensuring a seamless restoration of the model without disrupting the environment stability.

4. Optimization of Architectural Efficiency

- **Challenge:** Identifying a model configuration that delivers high precision while remaining computationally efficient for processing data.
- **Solution:** The **YOLOv8 Nano** variant was selected for its optimized architecture. It achieved a remarkable **mAP50 of 99.4%**, successfully balancing high precision with lightweight resource consumption.

9. Conclusion

In conclusion, the system was able to detect parking spaces and classify them into vacant and occupied using the YOLOv8 model with high accuracy, in addition to identifying the nearest available parking spot based on the user's location. The A* algorithm was also efficiently applied to generate the shortest possible path within the parking environment, enabling the system to provide a practical and clear recommendation to the user.

Experimental results also showed that the system operates accurately and effectively in different cases, where it achieved high performance in evaluation metrics such as accuracy, recall, and mAP, confirming the model's ability to distinguish parking spots with high precision. The routing algorithm also proved successful in selecting the optimal path and reducing the time and effort spent searching for a suitable parking spot, even under diverse environmental conditions, reflecting the system's strength in handling different real-world scenarios.

10. Future Work

The proposed system can be further developed in the future by adding several improvements and features that enhance its efficiency and improve the user experience. One of the possible enhancements is integrating live cameras instead of relying only on static images, which would allow real-time monitoring of parking spaces and continuous updates of their status. Another potential development is adding a data and statistics analysis system to study parking usage patterns and peak congestion times, which would support more efficient decision making. In addition, the system can be improved by integrating GPS services and navigation platforms such as Google Maps to provide end to end guidance from external roads to the exact parking spot, as well as developing mobile application support that enables users to view real time parking availability before arriving at the location.

Future work could also focus on improving the system's accuracy and scalability by supporting larger parking environments and enhancing vehicle detection performance under different lighting and weather conditions

Stage 4

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Task Distribution

Student Name	Initial task
Enas Abdullah Saud	Nearest Parking Recommendation Algorithm + Experimental Results and Discussion
Reema Salem Alharbi	responsible for reviewing and writing the related work section, searching past literature, and managing the overall data pre-processing and image preparation phase.
Rema Saed Althaqfi	responsible for developing and implementing the YOLOv8 core framework, conducting a comprehensive error analysis on the evaluation metrics, and identifying and documenting the technical project challenges.
Noran Abdullah Aljodi	Responsible for the Proposed Methodology and Implementation section, system architecture design, and user interface development for the smart parking detection and nearest parking recommendation system.
Asail Hamdan Al-Osaimi	Introduction (Overview, Problem Statement, Goals, Gaps ,Solution) + Future Work+Conclusion

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